

CSE 114 - Fundamentals of Computer Programming – 2026 Spring

Hw 1

For all parts: The interest rate is provided as an integer in percentage. Use double for amount. Return the interest value as a double value.

1) Write a function f1 which calculates simple monthly interest for a given principal amount and interest rate. The interest rate is provided as an integer in percentage. Use double for amount.

Simple interest is calculated using the formula,

$$\text{Simple Interest (SI)} = \text{Principal Amount} * (\text{Rate} / 100)$$

If you write the formula as $\text{Rate} / 100$, you will notice that the result is always zero. This is because, in C expressions also have types. Here is the relevant section from the textbook:

Data Type of an Expression

The data type of each variable must be specified in its declaration, but how does C determine the data type of an expression? The data type of an expression depends on the type(s) of its operands. Let's consider the types of expressions involving operands that are integers of type `int` or real numbers of type `double`.^{*} For example, the expression

`ace + bandage`

is type `int` if both `ace` and `bandage` are type `int`; otherwise, it is type `double`. In general, an expression of the form

`ace arithmetic_operator bandage`

is of type `int` if *both* `ace` and `bandage` are of type `int`; otherwise, it is of type `double`.

An expression that has operands of both type `int` and `double` is a **mixed-type expression**. The data type of such a mixed-type expression will be `double`.

Rate is `int`, 100 is `int` hence `Rate / 100` is `integer` also (you lose the fractional part, only the integral part is taken – not rounded, just what it is). So `40/100` is 0 while `40/100.0` is 0.4.

Either you can use `Rate / 100.0` or `(double) Rate / 100`. The second one is called “**explicit type casting of an expression**” and we will talk about it in the class. The value used for Rate inside this particular expression becomes double after this casting operation (ex. if it is 60, it becomes 60.0).

Think about this: `(double) ( Rate / 100 )` will not work, why?

2) Write a function f2 which calculates compound interest for a given principal amount and interest rate for **two months**.

You should find the compound interest as follows:

$$R1 = \text{Principal Amount} * (\text{Rate} / 100)$$

$$R2 = \text{New Amount} * (\text{Rate} / 100)$$

where `New Amount = Principal Amount + R1`.

But, you should not find R1 and R2 with these mathematical operations. You should find both of the by calling the function f1.

You should return R1 + R2.

3) Write a function f3 which calculates compound interest for a given principal amount and interest rate for **three months**.

You should find the compound interest as follows:

$$\begin{aligned} R1 &= \text{Principal Amount} * (\text{Rate} / 100) \\ R2 &= \text{New Amount}_1 * (\text{Rate} / 100) \\ R3 &= \text{New Amount}_2 * (\text{Rate} / 100) \end{aligned}$$

where $\text{New Amount}_1 = \text{Principal Amount} + R1$ and

$\text{New Amount}_2 = \text{Principal Amount} + R1 + R2$

But, you should not find R1 and R2 with these mathematical operations. You should find

R1 + R2 by calling f2 and

R3 by calling f1.

You should return R1 + R2 + R3.

4) Write the main function to test the above functions. Read the amount, interest rate and `num_months` from the user, find the interest value and display it to the user. `num_months` is the number of months and you can assume that the user will always enter 1, 2 or 3.

You will need to use if statements in main and we haven't studied them yet. Just use the following structure in your code:

```
if(num_months==1){
    // find the interest for 1 month here
}
else{
    if(num_months==2){
        // find the interest for 2 months here
    }
    else{
        // find the interest for 3 months here
    }
}
```

Submission – Write a single C program for all parts and submit your file with the given format:
`name_surname.c`